

KELOWNA, BC, Canada

WMO: 712030

Lat: 49.960N

Lon: 119.380W

Elev: 1421

StdP: 13.96

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	1.5	8.3	-5.4	4.3	3.4	1.1	6.1	10.0	18.9	29.1	16.3	26.9	3.3	0	0.548

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	30.1	91.8	64.4	88.4	63.3	84.9	62.1	66.4	85.9	64.7	83.7	63.2	81.4	7.0	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.8	80.8	70.6	57.9	75.4	69.3	56.1	70.6	68.4	31.7	85.7	30.4	83.4	29.3	81.9	74.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.6	15.4	12.8	DB	-5.5	97.3	8.7	2.7	-11.7	99.3	-16.8	100.9	-21.7	102.4	-28.0	104.4	
			WB	-6.2	70.4	8.7	2.4	-12.5	72.1	-17.5	73.5	-22.4	74.8	-28.7	76.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.8	27.9	29.8	38.9	46.5	55.2	61.6	68.3	66.7	57.4	44.9	35.2	27.8
	DBStd	15.89	9.00	8.29	6.93	5.51	6.10	5.60	5.25	5.43	5.84	5.98	7.98	8.51
	HDD50	3067	684	565	347	132	20	1	0	0	8	177	445	688
	HDD65	6917	1149	985	810	554	308	132	27	46	236	622	895	1153
	CDD50	1896	0	0	2	27	182	349	568	519	230	19	0	0
	CDD65	273	0	0	0	0	4	30	130	100	9	0	0	0
	CDH74	5123	0	0	0	7	226	658	2210	1742	279	1	0	0
	CDH80	2019	0	0	0	0	51	223	975	704	66	0	0	0
	WSAvg	4.2	3.8	3.7	4.6	5.0	5.0	4.9	4.2	3.8	3.5	3.6	3.6	3.6
	PrecAvg	18.3	1.9	1.2	1.2	1.1	1.8	2.2	1.2	1.1	1.2	1.7	1.8	2.0
Precipitation	PrecMax	23.0	2.9	2.2	1.8	2.3	3.8	4.2	4.4	2.7	3.0	2.9	3.7	3.2
	PrecMin	13.1	1.1	0.7	0.6	0.4	0.6	1.1	0.0	0.1	0.2	0.6	0.7	0.9
	PrecStd	2.6	0.5	0.4	0.3	0.5	0.7	0.9	0.9	0.7	0.7	0.6	0.7	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.6	49.7	62.7	74.2	85.6	92.1	97.0	94.6	86.7	68.4	55.5	47.1
		MCWB	39.7	41.4	48.4	55.5	62.2	64.6	66.0	64.9	62.2	55.8	48.2	40.9
	2%	DB	41.4	46.0	57.5	68.6	80.5	86.2	93.1	90.7	80.9	62.7	51.2	42.6
		MCWB	37.6	39.3	45.5	52.1	59.4	62.1	64.6	64.3	60.9	52.3	45.3	38.0
	5%	DB	39.1	42.8	54.1	64.0	75.4	81.3	89.6	87.6	76.8	59.3	48.3	39.6
		MCWB	36.2	37.8	43.6	49.4	57.0	60.3	63.7	63.4	59.2	50.1	43.3	36.3
	10%	DB	37.2	40.3	50.9	60.1	70.7	77.0	85.8	83.8	72.0	56.3	46.0	37.4
		MCWB	35.0	36.0	41.9	47.1	54.8	58.7	62.5	62.0	56.8	48.6	42.1	34.5
	0.4%	WB	40.7	43.2	49.4	56.7	63.4	66.6	69.1	68.5	65.1	57.4	49.9	42.1
		MCDB	44.4	47.3	59.6	72.0	81.9	85.5	90.7	88.6	80.0	64.9	53.8	45.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	38.2	40.3	46.7	53.4	60.3	63.7	66.7	66.0	62.4	54.3	46.8	39.0
		MCDB	40.9	44.2	55.6	65.9	76.1	82.5	87.1	85.0	77.4	60.2	49.6	41.5
	5%	WB	36.5	38.3	44.8	50.5	58.2	61.6	64.9	64.4	60.2	51.7	44.5	36.8
		MCDB	38.6	42.0	52.4	61.8	72.3	78.2	84.7	83.1	74.1	57.2	47.6	39.0
	10%	WB	35.1	36.5	42.7	48.2	56.2	59.7	63.4	62.9	58.3	49.5	42.3	34.9
		MCDB	36.9	39.8	49.7	58.1	68.1	74.0	82.3	81.0	70.2	55.1	45.1	36.9
Mean Daily Temperature Range	5% DB	MDBR	10.1	14.9	19.5	24.2	26.0	25.7	30.1	30.5	26.7	20.2	13.3	10.5
		MCDBR	11.6	16.1	24.4	32.5	35.8	35.7	38.4	37.7	35.3	24.6	16.0	12.1
		MCWBR	8.7	11.0	14.5	18.3	17.6	15.6	15.1	15.7	17.3	15.0	11.0	9.1
	5% WB	MCDBR	10.4	14.9	22.5	29.9	31.7	31.8	34.4	34.2	31.7	21.5	14.3	11.1
		MCWBR	7.9	10.5	13.9	17.3	16.1	14.6	14.4	15.0	16.3	13.9	10.3	8.8
Clear-Sky Solar Irradiance	taub		0.301	0.294	0.321	0.341	0.368	0.364	0.359	0.370	0.341	0.333	0.307	0.282
	taud		2.351	2.464	2.435	2.413	2.373	2.409	2.451	2.415	2.481	2.474	2.450	2.427
	Ebn at Noon		234	272	282	287	282	284	283	274	272	253	233	227
	Edh at Noon		21	24	30	34	36	36	34	33	28	24	19	17
All-Sky Solar Radiation	RadAvg		295	604	974	1430	1711	1800	2059	1757	1227	690	314	224
	RadStd		33	59	98	112	135	167	166	122	108	88	38	27

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+216	+65
(47)	Regional (19 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page